



Long-term risk of dementia following encephalitis: a large-scale retrospective cohort study of electronic health records

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Abstract

Background Encephalitis is an acute inflammation of the brain parenchyma, typically caused by microbial infections or autoimmune mechanisms, and represents a potentially life-threatening neurological condition affecting thousands of individuals each year in the United States. Although survivors may experience persistent neurological sequelae, long-term association of encephalitis and subsequent dementia and related diagnoses across different age groups and encephalitis etiologies remains poorly characterized. This study aimed to quantify the association between encephalitis and the risk of developing dementia and related diagnoses across age groups and encephalitis etiologies.

Methods We conducted a retrospective cohort study using TriNetX U.S. electronic health records. We first estimated the database specific cumulative incidence and prevalence of encephalitis. Patients diagnosed with encephalitis were then compared with propensity score matched controls for the incidence of a composite outcome of dementia and related diagnoses and its individual diagnostic categories, epilepsy (positive control) and mortality. Analyses were stratified by four age groups (< 20 years, 20–40 years, 40–60 years and > 60 years), and encephalitis etiological subtypes (non-infectious/post-infectious inflammatory, viral, bacterial, fungal/parasitic and unspecified). To assess specificity, the encephalitis cohort was additionally compared against three comparator cohorts (sepsis, meningitis and stroke). Risk ratios and risk differences were calculated after accounting for confounding variables.

Results The overall cumulative incidence and prevalence of encephalitis were 74.3 and 90.2 per 100,000 persons, respectively, during the 2014–2024 study period. Encephalitis was associated with an increased risk of composite of dementia and related diagnoses across all age groups. When the outcome was restricted to specific dementia codes, the association remained robust in older adults (40–60 years and > 60 years), but was no longer evident in younger cohorts, indicating that signal in younger individuals may reflect post-encephalitic neurological sequelae. Etiology specific analysis demonstrated that non-infectious/post-infectious inflammatory encephalitis conferred the highest relative risk for dementia, followed by viral, fungal/ parasitic and unspecified encephalitis, while bacterial encephalitis showed no significant association. Compared to sepsis, meningitis and stroke, the encephalitis cohort exhibited higher risk for the composite outcome and individual diagnostic categories. Vascular dementia risk was sole exception, being higher in the stroke cohort than in encephalitis cohort, consistent with known vascular pathogenesis of post-stroke dementia. Consistent with previous studies, epilepsy risk was significantly increased across all encephalitis cohorts. Mortality was also significantly higher in the encephalitis group across all comparisons except for sepsis.

Conclusions Encephalitis is associated with a markedly elevated long-term risk of subsequent dementia and related diagnoses in electronic health records data. These findings emphasize the need for long-term monitoring of encephalitis survivors regardless of age or etiology.

Keywords Encephalitis · Dementia · Neurodegeneration · Neuroinflammation · Electronic health records

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Introduction

Dementia is a devastating neurological syndrome characterized by progressive cognitive decline, memory loss and behavioral changes. Dementia remains one of the significant health challenges that affects millions of people worldwide and is projected to grow significantly due to aging populations [1, 2]. It is a major cause of disability among older adults and is associated with significant socio-economic burden with higher cost of health care and long-term care for dementia patients [3]. The most common cause of dementia is Alzheimer's disease that accounts for over 60% of dementia cases. Other cases of dementia include vascular dementia, Lewy body dementia and frontotemporal lobar dementia [1, 4]. Currently, there is no cure for dementia.

Dementia is a multifactorial syndrome involving several risk factors including aging, life style, environmental factors, comorbidities and genetics [5]. Although the precise mechanisms underlying the onset of dementia are not yet clear, inflammation has emerged as one of the central processes underlying its pathogenesis [6–8].

Inflammation is a defensive process crucial for host defense against injury and infection as well as homeostasis maintenance. However, dysregulated inflammation can be detrimental to health. In the brain, persistent activation of microglia and astrocytes triggers the release of pro-inflammatory cytokines, which can cause neuronal injury and synaptic dysfunction [7, 9, 10]. This sustained neuro-inflammatory response not only accelerates amyloid deposition and tau hyperphosphorylation but also interferes with neuronal repair and neurogenesis [11–14]. In addition, systemic inflammation, such as microbial infections or chronic systemic conditions like obesity and diabetes have been associated with increased levels of circulating cytokines that can exacerbate neurodegenerative processes by altering the permeability of blood–brain barrier and allowing bidirectional crosstalk between peripheral and central nervous system [15–23]. Collectively, these studies suggest inflammation as a potentially modifiable risk factor for neurodegeneration.

While chronic low grade neuroinflammation and systemic infections have been linked to dementia risk [24–28], the contribution of direct central nervous system inflammation to long-term dementia risk remains poorly characterized. Most studies have focused on gradual inflammatory processes associated with aging, systemic infections such as sepsis or chronic and latent viral infections such as human immunodeficiency virus and herpes simplex virus [20, 24, 25, 27, 29–37]. While the long-term cognitive impact of acute brain injury such as traumatic brain injury and stroke is well documented [38–45], and

the role of latent herpes virus infections in neurodegeneration has been investigated in recent large-scale cohorts [32, 46, 47], the longitudinal sequelae of acute, direct brain parenchymal inflammation, as occurs in encephalitis, is only beginning to be understood in large populations [26].

Encephalitis is characterized by rapid-onset (acute or subacute) severe inflammation of the brain parenchyma. It is caused by either infectious agents such as virus, bacteria or fungus or autoimmune process targeting neuronal or glial autoantigens [48, 49]. Encephalitis affects individuals across all age groups and while many patients recover, up to 30–60% experience persistent neurological sequelae, including cognitive deficits, epilepsy and behavioral disturbances [50, 51]. Several mechanisms suggest that encephalitis could lead to long-term dementia risk: it can cause direct brain injury to regions vulnerable to neurodegeneration [52–54], can disrupt blood–brain integrity [55] and particularly in viral and autoimmune forms, initial acute inflammatory events may trigger secondary, chronic neuroinflammation with persistent immune dysregulation [56, 57].

Despite the potential risk, the long-term incidence of dementia following encephalitis has not been systematically quantified across all etiologies. While recent large-scale studies using electronic health records and national biobanks have demonstrated a significant link between viral infections, including herpes simplex virus and viral encephalitis and subsequent neurodegeneration, evidence regarding autoimmune and other non-viral etiologies remains limited to small clinical cohorts or case reports [26, 46, 47, 50, 58–66]. However, it remains unclear whether dementia risk varies across distinct encephalitis etiologies (non-infectious/post-infectious inflammatory, viral, bacterial, parasitic and fungal), how it is influenced by patient age at onset, and how it compares to the risk following other acute neurological conditions. Furthermore, current post-encephalitis clinical care focuses primarily on acute recovery, without standardized protocols for long-term clinical follow-up. Ultimately, quantifying the frequency and risk factors for dementia after encephalitis would help identify high-risk patients who may benefit from long term cognitive monitoring and early intervention.

To address this, we conducted a large-scale retrospective cohort study using TriNetX US collaborative network electronic health records to: 1) estimate the incidence and prevalence of encephalitis and its subtypes within the database, 2) quantify the subsequent risk of dementia and related diagnoses among people with prior encephalitis compared to propensity score-matched controls, across age groups and individual diagnostic categories, 3) determine whether this risk varied by encephalitis etiology, and 4) contextualize these finding by comparing encephalitis survivors to patients with meningitis, sepsis and stroke. This study aims to determine whether acute brain inflammation from encephalitis

contributes to long-term dementia and neurocognitive risk and to inform clinical surveillance strategies for survivors.

Methods

Data source

We used data from TriNetX, a global federated research network platform, containing de-identified electronic health records from participating healthcare organizations (HCOs). This network includes longitudinal clinical data such as demographics, procedures, medications, diagnoses, laboratory values and vital signs from in-patient and out-patient encounters. TriNetX data is de-identified in compliance with Health Insurance Portability and Accountability Act and General Data Protection Regulation. Since patient data used in our study is de-identified, patient informed consent is waived as per guidelines of the institutional review board.

This study primarily analyzed data from TriNetX US collaborative network, which aggregates electronic health records from 72 HCOs. Patient cohorts were identified using international statistical classification of diseases and related health problems, tenth version (ICD-10) diagnostic codes, current procedural terminology (CPT) and RxNorm.

Study design

We conducted a retrospective cohort study using the TriNetX US collaborative network. This study comprised two analyses. In the first analysis, we utilized the platform's built in Incidence and Prevalence module to quantify frequency of encephalitis within the TriNetX database among individuals with at least one healthcare encounter (recorded as 'visit') during the study period. We used data from January 1, 2014 to December 31, 2024 for this analysis. Subgroup analysis was performed to estimate these metrics separately for encephalitis categorized based on etiological subtypes—1) non-infectious/post-infectious inflammatory encephalitis, 2) viral encephalitis, 3) bacterial encephalitis, 4) fungal/parasitic encephalitis and 5) unspecified encephalitis. The diagnostic ICD-10 codes to build these cohorts are listed in Supplementary Table 1.

In the second analysis, we determined the long-term risk of developing dementia and related diagnoses following a diagnosis of encephalitis. We identified patients diagnosed with encephalitis on or after January 1, 2005 up until the date of analysis (last day of analysis: June 20, 2026). We used a longer period for this analysis to capture more patients and allow adequate follow-up time. Patients with encephalitis were stratified by age into four groups 1) <20 years, 2) 20–40 years, 3) 40–60 years, and 4) >60 years and compared with an age-stratified control cohort (presented for a general

health examination without any complaints and encephalitis diagnosis) using TriNetX's built in analytics tools. To minimize confounding, individuals with a prior diagnosis of meningitis, myelitis, meningoencephalitis or intracranial injury were excluded from the study. Furthermore, to ensure all neurological and cognitive outcomes represented new events following encephalitis diagnosis, we excluded patients who had documented history of dementia or epilepsy prior to their index encephalitis diagnosis. Specifically, patients were excluded if they had a prior diagnosis of Alzheimer's disease (G30), other degenerative diseases of nervous system (G31), vascular dementia (F01), dementia in other diseases classified elsewhere (F02), unspecified dementia (F03), prior use of dementia medications (donepezil, galantamine, rivastigmine or memantine), or history of epilepsy and recurrent seizures (G40), at any time before cohort entry. Supplementary Table 1 list the diagnostic ICD-10 codes to build these cohorts. We assessed the risk within a 10-year follow-up period.

To assess the robustness of association, a series of sensitivity analyses were conducted. We repeated the primary analyses using alternative follow-up windows, including 5-year follow-up and anytime follow-up period after encephalitis diagnosis. We conducted landmark analyses excluding any dementia and related diagnoses outcomes occurring within 3 months, 6 months and 12 months post-encephalitis.

We also conducted analysis stratified by sex to assess if association between encephalitis and dementia differs between male and females.

Additionally, we repeated the primary analysis categorized by encephalitis subtypes, comparing each subtype to a matched control cohort composed of patients with similar underlying exposures but did not develop encephalitis. This approach was used to reduce confounding from underlying infections and immune conditions, and to estimate the direct association between encephalitis and dementia. For example, patients with viral, bacterial or fungal/parasitic encephalitis were each compared with control cohorts who had the corresponding viral, bacterial or fungal/parasitic infections without a diagnosis of encephalitis. The unspecified encephalitis cohort consists of cases where the underlying etiology could not be determined. Because patients with encephalitis frequently present with systemic symptoms such as seizures, high fever and require hospitalization, we used a control cohort consisting of patients who were hospitalized with high fever but without encephalitis diagnosis, to account for the effect of acute illness and hospitalization. The non-infectious/post-infectious inflammatory encephalitis cohort comprised of autoimmune (such as anti-NMDA receptor encephalitis and Hashimoto's encephalopathy), post-infectious encephalitis, including acute disseminated encephalitis and encephalomyelitis (ADEM) and acute necrotizing hemorrhagic encephalopathy. For this subtype, we used a control

cohort with autoimmune conditions (systemic lupus erythematosus, rheumatoid arthritis or autoimmune thyroiditis) and relevant infectious viral triggers (varicella, measles, mumps, influenza, or infectious mononucleosis) that are associated with encephalitis risk but who did not develop encephalitis, allowing us to distinguish the impact of immune-mediated brain inflammation from the general effect of underlying autoimmune or infectious exposures. The diagnostic ICD-10 codes to build these cohorts are listed in Supplementary Table 1.

Finally, comparative analyses were performed between encephalitis and other neurologically relevant cohorts including meningitis, stroke and sepsis to contextualize the magnitude of risk of dementia across acute neurological and systemic inflammatory conditions. Supplementary Table 1 lists the diagnostic ICD-10 codes to build these cohorts.

Outcomes measured

The outcomes were the new onset of neurological or neurodegenerative conditions following a diagnosis of encephalitis. The primary outcome of interest was dementia and related diagnoses defined as composite of Alzheimer disease (G30), other degenerative diseases of nervous system (G31), vascular dementia (F01), dementia in other diseases classified elsewhere (F02) or unspecified dementia (F03). These dementia and related diagnoses (ICD-10 G30, G31, F01, F02 and F03) were also analyzed separately. Secondary outcomes were epilepsy and recurrent seizures (G40) and all-cause mortality. All subtypes, etiology and comparator cohorts' analyses were designated as secondary and exploratory analyses. We assessed the risk of dementia within a 10-year follow-up period after the index encephalitis diagnosis.

Propensity score matching analysis

To minimize the impact of confounding variables between encephalitis cohort and control cohorts, as well as across all other comparison cohorts, propensity score matching was performed within TriNetX platform. Patients were matched 1:1 between cohorts based on propensity scores using nearest-neighbor algorithm without replacement that uses logistic regression to generate same size cohorts. A default caliper of 0.10 standard deviation was applied, and cohorts were randomly shuffled prior to matching to reduce ordering bias. Covariates included in the propensity score matching were known or suspected confounders for dementia: age at index, sex, race, ethnicity, overweight and obesity (E66), tobacco use (Z72.0), mental and behavioral disorders due to psychoactive substance use (F10-F19), Diabetes mellitus (E08-E13), chronic kidney disease (N18), heart diseases (I30-I5A), hypertensive diseases (I10-I1A), ischemic heart

diseases (I20-I25) and cerebral infarction (I63). Following matching, a standardized mean difference (SMD) of less than 0.1 was considered indicative of adequate balance for covariates between the cohorts. The baseline characteristics of all cohort comparisons before and after propensity score matching is detailed in Supplementary Table 2–4.

Statistical analysis

All analyses were performed using built in analytical tools of the TriNetX platform. Incidence proportion was calculated as the number of new encephalitis cases during the study period (2014–2024) divided by the total number of individuals with at least one healthcare encounter during the same period. A 12 month of look-back period was applied to ensure exclusion of existing cases and to capture only new events. Prevalence was calculated as total number of patients with encephalitis at a specified period of time divided by the total number of patients in the network at that same time. Both incidence and prevalence were expressed as cases per 100,000 persons.

For dementia outcomes, risk ratios (RRs) and risk difference (RDs) were calculated and reported with 95% confidence intervals (CIs). Statistical significance was defined as a 95% CI that did not include 1.0 for RRs. These estimates were visualized using forest plots, where statistical significance is graphically indicated by CI error bars that do not cross the vertical line of no effect. For categorical outcomes, statistical significance was determined using chi-square tests. Kaplan–Meier survival analyses were used to estimate survival probabilities, and survival curves between cohorts were compared using log-rank test. Hazard ratios (HRs) with 95% confidence interval were estimated using Cox-proportional hazard models. To address mortality as a competing risk for dementia and related outcomes, a competing risk analysis treating death and dementia as competing events was performed using the Competing risk analytical tool available within the TriNetX platform on unmatched cohorts. Graphical visualizations were generated using R statistical software.

Results

Incidence and prevalence of encephalitis

To quantify the burden of encephalitis, we utilized the built-in Incidence and Prevalence analytical tool within the TriNetX database. We established a base cohort of ~129,000,000 patients using search term 'visit' in the TriNetX network. The overall cumulative incidence for encephalitis during the study period from January 1, 2014, to December 31, 2024, was 74.3 per 100,000 persons. The

overall cumulative prevalence for encephalitis during the study period was 90.2 per 100,000 persons. Both incidence proportion and prevalence of encephalitis varied by age and sex during the study period 2014–2024. The cumulative incidence of encephalitis increased with age, with the highest burden observed in age group 65–69 years (125.7 per 100,000 persons) and the lowest in age group

15–19 years (44.2 per 100,000 persons) (Fig. 1A). The cumulative incidence was slightly higher among females (80.8 per 100,000 persons) than among males (69.0 per 100,000 persons) (Fig. 1B). A similar age and sex specific pattern were observed for prevalence, with higher prevalence in older age groups (Fig. 1D) and among females (Fig. 1E).

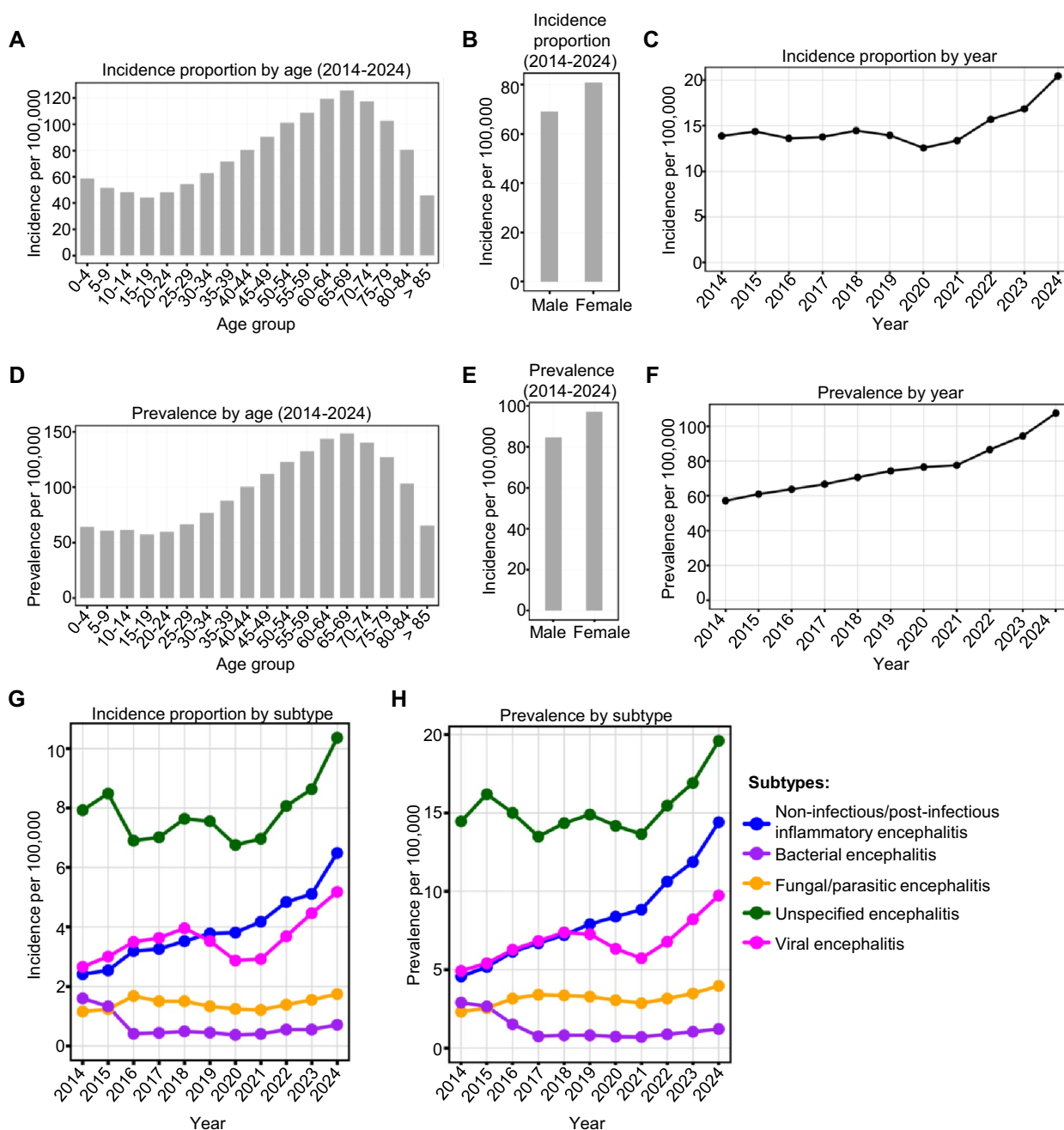


Fig. 1 Incidence and prevalence of encephalitis (2014–2024): Cumulative incidence of encephalitis from 2014–2024 are shown stratified by **A** age groups and **B** sex. **C** Annual incidence of encephalitis cases from 2014–2024 to illustrate temporal trends over the study period. Incidence is reported as cases per 100,000 persons. Prevalence of

encephalitis from 2014–2024 are shown stratified by **D** age groups and **E** sex. **F** Temporal trends in annual prevalence of encephalitis cases from 2014–2024. Prevalence is reported as cases per 100,000 persons. **G** Incidence and **(H)** prevalence of encephalitis by etiological subtype over the study period 2014–2024

Annual temporal analysis from 2014–2024 showed that the incidence of encephalitis remained relatively stable between 2014 and 2019 followed by a slight decline in 2019–2021 and then a steady increase through 2024, where it reached the highest levels (20.4 per 100,000 persons) (Fig. 1C). Prevalence showed a steady increase over time, with the highest value observed in 2024 (107.4 per 100,000 persons), representing a notable increase from 2014 (57.1 per 100,000 persons) (Fig. 1F).

We further stratified the encephalitis into five groups based on etiology to characterize the distribution of encephalitis by etiological subtypes. These five groups are: 1) non-infectious/post-infectious inflammatory encephalitis 2) viral encephalitis 3) bacterial encephalitis 4) fungal/parasitic encephalitis, and 5) unspecified encephalitis. The diagnostic ICD-10 codes to build these cohorts are listed in Supplementary Table 1. Across the 2014–2024 study period, when stratified by etiology, unspecified encephalitis showed the highest incidence and prevalence followed by non-infectious/post-infectious inflammatory encephalitis, viral encephalitis, and fungal/parasitic encephalitis, while bacterial encephalitis remained least frequent subtype (Fig. 1G–H). A transient decline in viral encephalitis and unspecified encephalitis cases was seen during 2019–2021, followed by an increase through 2022, paralleling the overall encephalitis trends (Fig. 1G–H). Non-infectious/post-infectious inflammatory encephalitis displayed gradual steady increase over the years (Fig. 1G–H). However, bacterial and fungal/parasitic encephalitis showed minimal temporal variation and remained consistently low over time (Fig. 1G–H).

Risk of subsequent dementia and other outcomes following encephalitis

Study population: There were ~129,000,000 patients in the TriNetX US collaborative network at the time of study analysis. To capture a large cohort of encephalitis cases and allow for a longer follow-up time for dementia outcome, the study period included all patients diagnosed with encephalitis on or after January 1, 2005 up until the date of analysis. We stratified patients with encephalitis into four age groups (1) less than 20 years, 2) 20–40 years, 3) 40–60 years, and 4) more than 60 years) and compared their risk of subsequent neurological outcomes and mortality to matched controls. Supplementary Fig. 1 is a layout of cohort composition. After propensity score matching analysis, cohorts were well-balanced (Table 1). After matching, the following patients remained: age group > 60 years ($n = 16,907$), age group 40–60 years ($n = 14,862$), age group 20–40 years ($n = 10,606$), and age group < 20 years ($n = 11,004$). The detailed baseline characteristics of each cohort before and after propensity score matching are outlined in Supplementary Table 2.

Age-stratified risk of dementia and other outcomes after encephalitis diagnosis:

Dementia risk

Overall, we find that risk of dementia and related diagnoses was significantly elevated in patients who experienced encephalitis across all age groups (Fig. 2A). The risk ratio (RR 95% CI) for dementia and related diagnoses for a 10-year follow-up was highest for age group < 20 years RR 95% CI 23.0 (13.952, 37.914) followed by age group 20–40 years RR 95% CI 14.28 (9.989, 20.416), age group 40–60 years RR 95% CI 5.164 (4.457, 5.982) and lowest in age group > 60 years 2.11 (1.982, 2.253) (Fig. 2A). While the relative risk was higher in younger population, the absolute increase in dementia and related diagnoses risk was greater in older individuals compared to younger individuals. Age group > 60 years risk difference: 0.081 (0.075, 0.088), age group 40–60 years risk difference: 0.058 (0.053, 0.062), age group 20–40 years risk difference: 0.040 (0.036, 0.044), and age group < 20 years risk difference: 0.031 (0.028, 0.035) (Fig. 2A).

The Kaplan Meier survival analysis further supported these findings (Fig. 2B–E). The difference in time to event trajectories between two groups was statistically significant across all age groups. The hazard ratios (HR 95%CI) for a 10-year follow-up for age group > 60 years was HR: 3.26 (3.05, 3.49) (Fig. 2B), age group 40–60 years HR: 6.74 (5.81, 7.83) (Fig. 2C), age group 20–40 years HR: 16.11 (11.26, 23.06) (Fig. 2D), and age group < 20 years HR: 29.32 (17.7, 48.40) (Fig. 2E).

Sensitivity analyses

We also assessed the risk of dementia and related diagnoses following an encephalitis diagnosis over a 5-year follow-up period and across the entire study period (anytime follow-up). We observed consistent results with the main analysis, with patients who had encephalitis exhibited a higher risk of developing dementia and related diagnoses (Supplementary Fig. 2A–B). Additionally, no significant differences in dementia and related diagnoses risk between male and female patients with encephalitis were observed except for age group 40–60 years where female cohort had slight lower risk compared to male cohort (Supplementary Fig. 3). We next conducted landmark sensitivity analyses excluding outcomes diagnosed with in 3 months, 6 months and 1 year. The risk remained elevated for all age groups except for < 20 age groups (Supplementary Fig. 4A–C).

In the initial analysis, we determined the risk of dementia and related diagnoses among patients with encephalitis. We next asked if risk of dementia differs among various age groups according to specific dementia and related diagnoses.

Table 1 Baseline characteristics of patients with encephalitis compared to controls after 1:1 propensity score matching

Characteristic	Age group >60 years			Age group 40-60 years			Age group 20-40 years			Age group <20 years		
	Encephalitis (n=16907)	Control (n=16907)	Standardized mean difference	Encephalitis (n=14862)	Control (n=14862)	Standardized mean difference	Encephalitis (n=10606)	Control (n=10606)	Standardized mean difference	Encephalitis (n=11004)	Control (n=11004)	Standardized mean difference
Demographics (%)												
Age at Index, mean(SD)	69.53(6.78)	69.54(6.79)	0.002692099	50.51(6.19)	50.52(6.21)	0.001561418	30.19(6.18)	30.22 (6.18)	0.00530364	9.59 (6.255)	9.60(6.25)	0.003818563
Male	8624 (51.01%)	8652 (51.17%)	0.003313031	7082 (47.65%)	7136 (48.02%)	0.007273735	4532 (42.73%)	4551 (42.91%)	0.003620405	5672 (51.55%)	5683 (51.65%)	0.002000292
Female	8273 (48.93%)	8245 (48.77%)	0.003313119	7774 (52.31%)	7720 (51.95%)	0.007273483	6064 (57.18%)	6044 (56.99%)	0.003809855	5324 (48.38%)	5313 (48.28%)	0.002000387
White	12568 (74.34%)	12581 (74.41%)	0.001761282	9098 (61.22%)	9124 (61.39%)	0.00359186	5851 (55.17%)	5854 (55.20%)	0.000568779	6201 (56.35%)	6207 (56.41%)	0.001099499
Black or African American	2033 (12.03%)	2036 (12.04%)	0.000545382	2802 (18.85%)	2831 (19.05%)	0.004978885	2126 (20.05%)	2148 (20.25%)	0.005171372	1695 (15.40%)	1693 (15.39%)	0.000503615
Other Race	622 (3.68%)	603 (3.57%)	0.006014257	964 (6.49%)	938 (6.31%)	0.007148365	814 (7.68%)	804 (7.58%)	0.003552057	921 (8.37%)	918 (8.34%)	0.000985186
Asian	564 (3.34%)	556 (3.29%)	0.002644097	507 (3.41%)	496 (3.34%)	0.004098957	431 (4.06%)	420 (3.96%)	0.005285181	487 (4.43%)	486 (4.42%)	0.000442081
Hispanic or Latino	1185 (7.01%)	1150 (6.80%)	0.008164842	2141 (14.41%)	2153 (14.49%)	0.002296716	1800 (16.97%)	1811 (17.08%)	0.002759565	1710 (15.54%)	1705 (15.49%)	0.00125496
Not Hispanic or Latino	12787 (75.63%)	12818 (75.82%)	0.004276468	10105 (67.99%)	10146 (68.27%)	0.005920369	6937 (65.41%)	6946 (65.49%)	0.001784468	7352 (66.81%)	7360 (66.89%)	0.001544336
Diagnosis (%)												
Hyper-tensive diseases (I10-I1A)	8632 (51.06%)	8646 (51.14%)	0.001656518	4692 (31.57%)	4739 (31.89%)	0.006794826	1281 (12.08%)	1310 (12.35%)	0.008350194	364 (3.31%)	373 (3.39%)	0.004546173
Other forms of heart disease (I30-I5A)	5839 (34.54%)	4885 (28.89%)	0.12147519	2986 (20.09%)	2098 (14.12%)	0.15918094	1325 (12.49%)	750 (7.07%)	0.18325989	610 (5.54%)	323 (2.94%)	0.12971757
Diabetes mellitus (E08-E13)	4377 (25.89%)	4374 (25.87%)	0.000405141	2362 (15.89%)	2370 (15.95%)	0.001471286	481 (4.54%)	508 (4.79%)	0.012074822	94 (0.85%)	100 (0.91%)	0.005833304
Ischemic heart diseases (I20-I25)	3600 (21.29%)	3597 (21.28%)	0.000433507	1292 (8.69%)	1277 (8.59%)	0.003591824	228 (2.15%)	221 (2.08%)	0.004585228	13 (0.12%)	13 (0.12%)	0

Table 1 (continued)

Encephalitis vs control cohorts											
Characteristic	Age group >60 years			Age group 40-60 years			Age group 20-40 years			Age group <20 years	
	Encephalitis (n=16907)	Control (n=16907)	Standardized mean difference	Encephalitis (n=14862)	Control (n=14862)	Standardized mean difference	Encephalitis (n=10606)	Control (n=10606)	Standardized mean difference	Encephalitis (n=11004)	Control (n=11004)
Overweight and obesity (E66)	2909 (17.21%)	2899 (17.15%)	0.001568163	2071 (13.94%)	2078 (13.98%)	0.001359091	941 (8.87%)	960 (9.05%)	0.006271801	323 (2.94%)	315 (2.86%)
Chronic kidney disease (N18)	2871 (16.98%)	2869 (16.97%)	0.000315103	1380 (9.29%)	1379 (9.28%)	0.000231875	478 (4.51%)	493 (4.65%)	0.006767031	91 (0.83%)	100 (0.91%)
Mental and behavioral disorders due to psychoactive substance use (F10-F19)	2580 (15.26%)	2554 (15.11%)	0.004285355	2786 (18.75%)	2805 (18.87%)	0.003271401	1489 (14.04%)	1494 (14.09%)	0.001356101	141 (1.28%)	135 (1.23%)
Cerebral infarction (I63)	1485 (8.78%)	1478 (8.74%)	0.001464292	812 (5.46%)	783 (5.27%)	0.008659137	265 (2.50%)	235 (2.22%)	0.018645503	88 (0.80%)	66 (0.60%)
Tobacco use (Z72.0)	474 (2.80%)	442 (2.61%)	0.011658827	503 (3.38%)	513 (3.45%)	0.003703245	204 (1.92%)	198 (1.87%)	0.004148901	10 (0.09%)	10 (0.09%)
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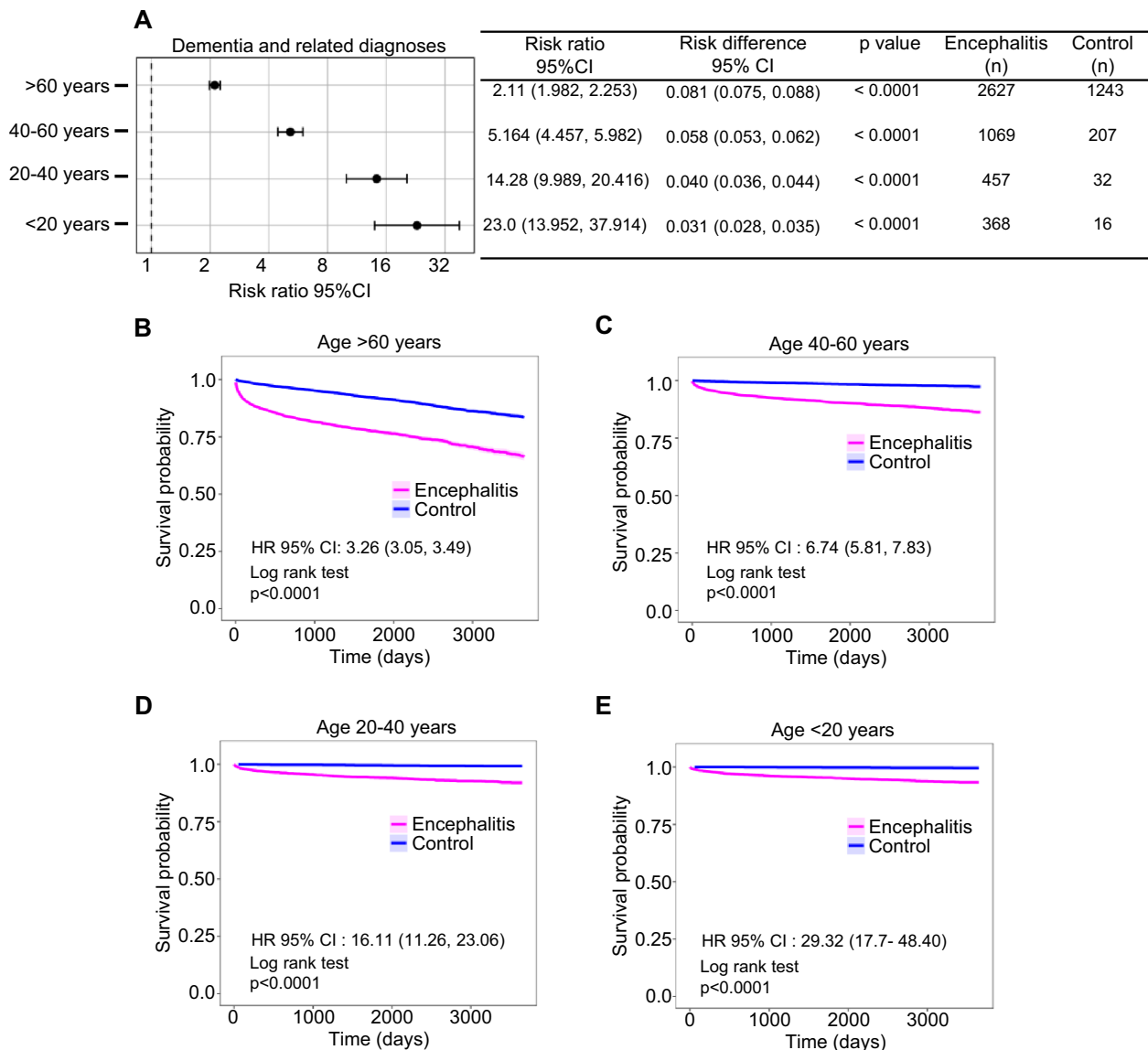


Fig. 2 Risk of dementia and related diagnoses following encephalitis. **A** Forest plot showing risk ratios and 95% confidence intervals (RR 95% CI) for dementia and related diagnoses among encephalitis patients compared to matched control cohorts. The encephalitis and control cohorts are stratified into four groups based on age. Dashed vertical line indicates value of risk ratio=1, where there is no significant association. The table lists risk ratios with 95% CI, risk differ-

ences with 95% CI, p value and number of dementia patients in each cohort. Kaplan Meier analysis for dementia and related diagnoses free survival among patients with encephalitis and matched control cohorts stratified by age group groups **B**> 60 years, **C** 40–60 years, **D** 20–40 years and **(E)**<20 years. Cox proportional hazard model was used to estimate hazard ratio with 95% CI

For this, we stratified outcomes by specific diagnostic categories using ICD-10 diagnostic codes. G30: Alzheimer's disease, G31: Other degenerative diseases of nervous system, F01: Vascular dementia, F02: Dementia in other diseases and F03: Unspecified dementia.

Older age groups (> 60 years and 40–60 years) consistently showed a significant increased risk for all individual diagnostic categories (Fig. 3A, B). In individuals aged > 60 years, RR 95%CI were as follows: Alzheimer's disease, 1.432(1.207, 1.698); other degenerative diseases

of nervous system, 2.152 (1.954, 2.371); vascular dementia, 2.148 (1.771, 2.605); dementia in other diseases, 2.245 (1.956, 2.576); and unspecified dementia, 2.4 (2.176, 2.648) (Fig. 3A). For individuals aged 40–60 years, RR 95% CI 95%CI were as follows: Alzheimer's disease, 3 (1.602, 5.617); other degenerative diseases of nervous system, 5.524 (4.494, 6.79); vascular dementia, 5.0 (2.972, 8.412); dementia in other diseases, 11.529 (7.028, 18.913); and unspecified dementia, 7.415 (5.356, 10.264) (Fig. 3B).

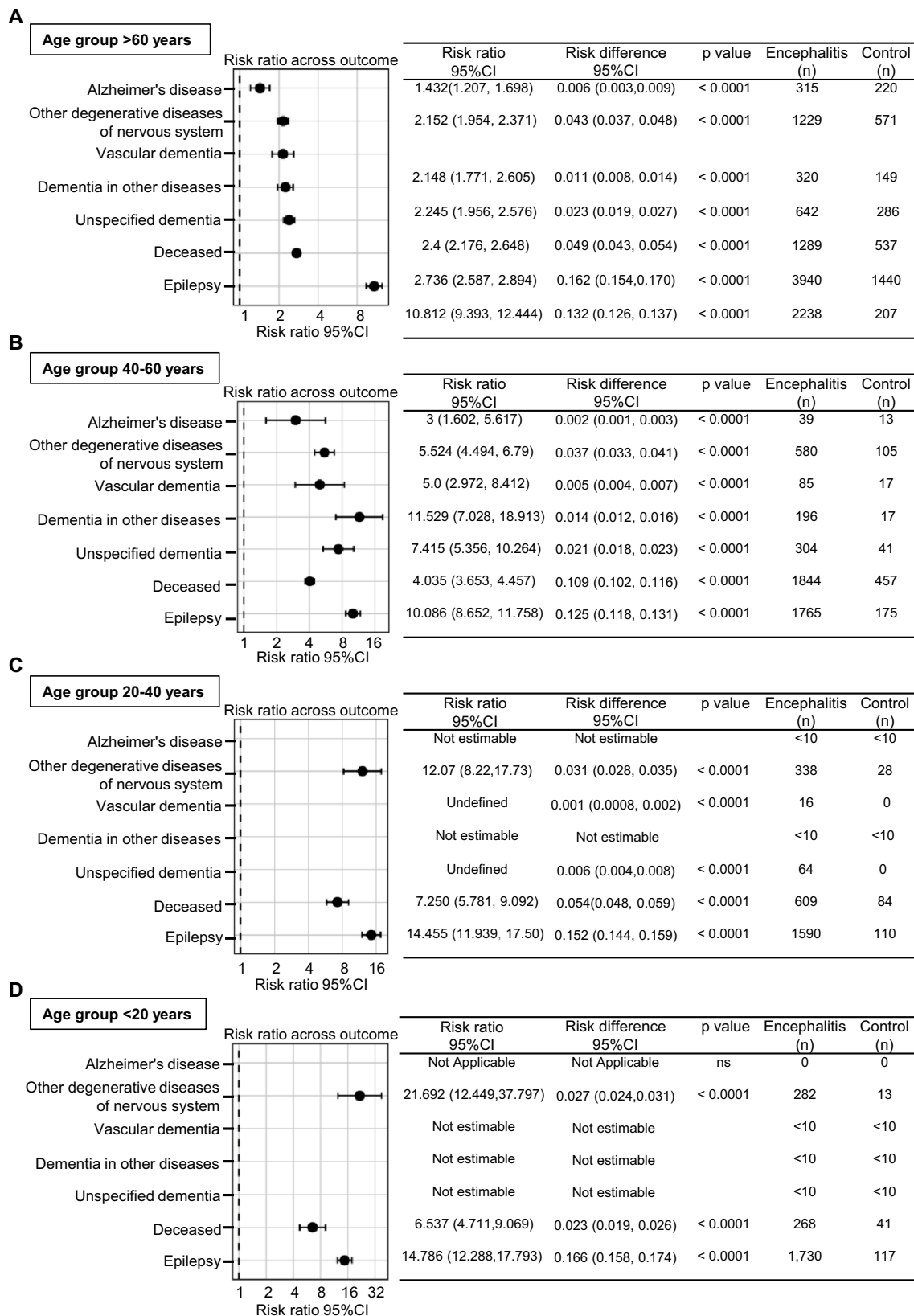


Fig. 3 Risk across individual diagnostic categories of dementia and related diagnoses following encephalitis. Forest plot showing risk of individual diagnostic categories (Alzheimer disease (G30), other degenerative diseases of nervous system (G31), vascular dementia (F01), dementia in other diseases classified elsewhere (F02) or unspecified dementia (F03) and secondary outcomes) following encephalitis. Risk ratios and 95% confidence intervals (RR 95% CI) are shown for patients stratified by age groups **A** > 60 years, **B** 40–60 years, **C** 20–40 years and **D** < 20 years. The table lists risk ratios with 95% CI, risk differences with 95% CI, p value and number of dementia patients in each cohort

In younger populations (< 20 years and 20–40 years), absolute numbers of individual diagnostic categories were low. (Fig. 3C, D). In individuals aged 20–40 years, RR 95%CI were as follows: Alzheimer's disease, not estimable due to fewer than 10 patients; other degenerative diseases of nervous system, 12.07 (8.22, 17.73); vascular dementia, undefined (0 patients in control cohort versus 16 in encephalitis cohort); dementia in other diseases, not estimable due to fewer than 10 patients; and unspecified dementia, Undefined (0 patients in control cohort versus 64 in encephalitis cohort) (Fig. 3C). In individuals aged < 20 years, recorded cases of Alzheimer's disease were 0 in both control and encephalitis cohort, and fewer than 10 cases were observed for vascular dementia, dementia in other diseases and unspecified dementia. However, RR 95%CI for other degenerative diseases of nervous system was 21.692 (12.449, 37.797) (Fig. 3D). Since in both younger cohorts (< 20 years and 20–40 years), signal appeared to be driven almost entirely by other degenerative diseases of nervous system (G31), we further conducted a sensitivity analysis where we measured outcome for specific dementia codes (Alzheimer disease (G30), vascular dementia (F01), dementia in other diseases classified elsewhere (F02) or unspecified dementia (F03). As expected, the risk was not estimable in the < 20 years and 20–40 years group after exclusion of other degenerative diseases of nervous system (G31) (Supplementary Fig. 5). However, the risk of dementia remained significantly higher in > 60 years (RR 95%CI 2.467 (2.262, 2.691)) and 40–60 years (RR 95%CI 6.088 (4.812, 7.702)) age group (Supplementary Fig. 5). These findings suggest that the overall burden of dementia and related diagnoses is concentrated in older patients, and the signal in younger populations likely reflects long term post-encephalitic neurological sequelae rather than dementia.

Mortality and epilepsy risk

Across all age groups, patients in the encephalitis cohorts experienced higher all-cause mortality compared to matched controls. The RR 95%CI for 10-year follow-up period were as follows: Age group > 60 years, 2.736 (2.587, 2.894); 40–60 years, 4.035 (3.653, 4.457); 20–40 years,

7.250 (5.781, 9.092); and < 20 years, 6.537 (4.711, 9.069) (Fig. 3A–D).

Because mortality represents a significant competing risk for long-term dementia, we conducted two analyses. First to observe the absolute risk, we utilized the Aalen-Johansen estimator (Competing risk built in TriNetX analytical tool) on the unmatched, descriptive cohorts to calculate cumulative incidence function treating death and dementia and related diagnoses as competing events. We observed that despite the higher mortality rate in encephalitis cohort, the cumulative incidence of dementia remained substantially higher (Supplementary Fig. 6A). Second, we evaluated a composite endpoint of dementia and related diagnoses or death within the propensity matched cohorts. The risk of this composite outcome remained significantly higher in encephalitis cohort compared to matched controls across all age groups (Supplementary Fig. 6B). The risk ratios are as follows: Age group > 60 years, 2.441 (2.337, 2.549); 40–60 years, 4.222 (3.882, 4.591); 20–40 years, 8.405 (6.919, 10.211); and < 20 years, 15.122 (11.038, 20.717). These analyses confirm that the heightened dementia risk is not solely driven by differential mortality between encephalitis and control cohorts. Epilepsy is a known neurological complication of encephalitis [67, 68]. We used this as a positive control outcome to confirm the validity of our cohorts. Consistent to previous reports, we saw a strong risk of epilepsy and recurrent seizures across all age groups following encephalitis diagnosis. The RR 95%CI were as follows: Age group > 60 years, 10.812 (9.393, 12.444); 40–60 years, 10.086 (8.652, 11.758); 20–40 years, 14.455 (11.939, 17.50); and < 20 years, 14.786 (12.288, 17.793) (Fig. 3A–D).

Risk of subsequent dementia and other outcomes by encephalitis subgroups

To assess if subsequent risk of dementia was driven by a particular etiology of encephalitis, we determined the risk of these outcomes across five encephalitis subtypes. For this analysis instead of using general population control, each encephalitis subtype cohort was compared with a matched group exposed to same underlying exposure but who did not develop encephalitis. The diagnostic ICD-10 codes to build these cohorts are listed in Supplementary Table 1. For example, patients with viral, bacterial or fungal/parasitic encephalitis were compared with individuals who had corresponding infectious triggers but did not develop encephalitis. Similarly, patients with non-infectious/post-infectious inflammatory encephalitis were compared with individuals with autoimmune conditions and/or relevant infectious triggers. This approach allowed us to isolate the effect of encephalitis itself from effects of underlying infections or immune challenge. The detailed baseline characteristics of each cohort before and after propensity score matching

are outlined in Supplementary Table 3. After matching, the following numbers of patients remained in each subgroup: non-infectious/post-infectious inflammatory encephalitis (n=9557), viral encephalitis (n=10,208), fungal/parasitic encephalitis (n=5100), bacterial encephalitis (n=1849) and unspecified encephalitis (n=13,103).

Dementia risk

Patients diagnosed with non-infectious/post-infectious inflammatory encephalitis showed the highest overall risk

of developing dementia and related diagnoses followed by viral encephalitis, fungal/parasitic encephalitis and unspecified encephalitis (Fig. 4A). Bacterial encephalitis did not confer any significant risk of dementia and related diagnoses compared to control cohort (Fig. 4A). The RR 95% CI are as follows: non-infectious/post-infectious inflammatory encephalitis, 3.928 (3.345, 4.613); viral encephalitis, 1.99 (1.768, 2.239); fungal/parasitic encephalitis, 1.622 (1.334, 1.972); unspecified encephalitis, 1.417 (1.298, 1.546); and bacterial encephalitis, 1.35 (0.973, 1.873) (Fig. 4A).

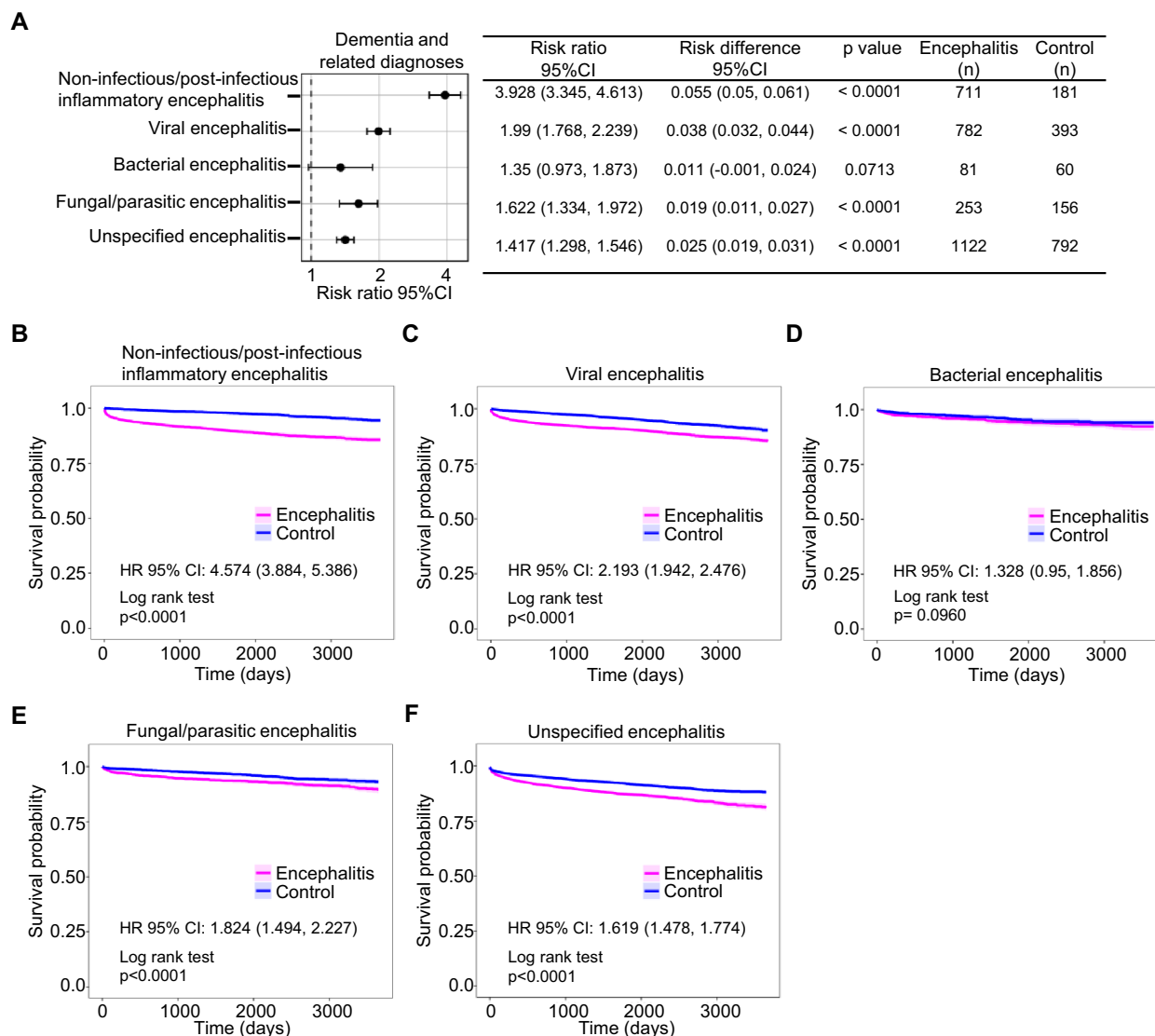


Fig. 4 Risk of dementia and related diagnoses following encephalitis stratified by subtype. **A** Forest plot showing risk ratios and 95% confidence intervals (RR 95% CI) for dementia and related diagnoses calculated for each encephalitis subtype compared to matched control cohorts. Dashed vertical line indicates value of risk ratio=1, where there is no significant association. The table lists risk ratios with 95% CI, risk differences with 95% CI, p value and number of dementia

patients in each cohort. Kaplan Meier analysis for dementia free survival among patients with encephalitis and matched control cohorts categorized by encephalitis subtype. **B** Non-infectious/post-infectious inflammatory, **C** viral, **D** bacterial, **E** fungal/parasitic, and **F** unspecified. Cox proportional hazard model was used to estimate hazard ratio with 95% CI

These findings were further supported by Kaplan Meier survival analysis (Fig. 4B–F). The difference in time to event trajectories between two groups was statistically significant across all subtypes except for bacterial encephalitis. The HR 95% CI were as follows: non-infectious/post-infectious inflammatory encephalitis, 4.574 (3.884, 5.386) (Fig. 4B); viral encephalitis, 2.193 (1.942, 2.476) (Fig. 4C); bacterial encephalitis, 1.328 (0.95, 1.856) (Fig. 4D); fungal/parasitic encephalitis, 1.824 (1.494, 2.227) (Fig. 4E); and unspecified encephalitis, 1.619 (1.478, 1.774) (Fig. 4F).

We next assessed the risk of developing specific individual diagnostic categories among various encephalitis subtypes. Non-infectious/post-infectious inflammatory encephalitis and viral encephalitis showed significantly higher risk of developing all individual diagnostic categories compared to matched controls (Supplementary Fig. 7A–B). For non-infectious/post-infectious inflammatory encephalitis cohort, RR 95%CI were as follows: Alzheimer's disease, 1.526 (1.015, 2.295); other degenerative diseases of nervous system, 4.333 (3.477, 5.4); vascular dementia, 2.433 (1.592, 3.719); dementia in other diseases, 3.884 (2.781, 5.424); and unspecified dementia, 3.819 (3.004, 4.855) (Supplementary Fig. 7A). For viral encephalitis, RR 95%CI were as follows: Alzheimer's disease, 1.395 (1.04, 1.87); other degenerative diseases of nervous system, 1.729 (1.461, 2.048); vascular dementia, 1.827 (1.304, 2.559); dementia in other diseases, 1.899 (1.491, 2.419); and unspecified dementia, 2.077 (1.764, 2.445) (Supplementary Fig. 7B).

Fungal/parasitic encephalitis subtype showed significant risk for all individual diagnostic categories except for Alzheimer's disease and vascular dementia (Supplementary Fig. 7D). The RR 95%CI were as follows: Alzheimer's disease, 0.647 (0.303, 1.38); other degenerative diseases of nervous system, 1.659 (1.271, 2.165); vascular dementia, 0.917 (0.515, 1.633); dementia in other diseases, 1.575 (1.062, 2.336); and unspecified dementia, 1.597 (1.165, 2.188) (Supplementary Fig. 7D).

Unspecified encephalitis subtype also showed significant risk all individual diagnostic categories except for Alzheimer's disease (Supplementary Fig. 8). The RR 95%CI were as follows: Alzheimer's disease, 0.985 (0.776, 1.25); other degenerative diseases of nervous system, 1.799 (1.576, 2.055); vascular dementia, 1.449 (1.121, 1.872); dementia in other diseases, 1.324 (1.105, 1.585); and unspecified dementia, 1.216 (1.077, 1.373) (Supplementary Fig. 8).

In contrast, bacterial encephalitis was the only subtype in this study where we did not observe any significant association between dementia outcome and encephalitis (Supplementary Fig. 7C). The RR 95%CI were as follows: Alzheimer's disease, 1.1 (0.468, 2.584); other degenerative diseases of nervous system, 1.125 (0.702, 1.803); vascular dementia, not estimable due to fewer than 10 patients; dementia in other diseases, 1.727 (0.824, 3.619); and

unspecified dementia, 1.556 (0.963, 2.512) (Supplementary Fig. 7C).

Mortality and epilepsy risk

All subtypes showed higher risk of all-cause mortality except for unspecified encephalitis (Supplementary Figs. 7A–D and 8). The RR 95% CI are as follows: non-infectious/post-infectious inflammatory encephalitis, 2.503 (2.23, 2.808); viral encephalitis, 2.377 (2.19, 2.581); fungal/parasitic encephalitis, 1.241 (1.118, 1.378); unspecified encephalitis, 0.958 (0.91, 1.009); and bacterial encephalitis, 1.951 (1.635, 2.329) (Supplementary Fig. 7A–D and 8).

Consistent with overall cohort analysis, all subtypes showed high risk of epilepsy (Supplementary Fig. 7A–D and 8). The RR 95% CI are as follows: non-infectious/post-infectious inflammatory encephalitis, 9.641 (7.987, 11.638); viral encephalitis, 11.886 (9.758, 14.478); fungal/parasitic encephalitis, 3.554 (3.022, 4.179); unspecified encephalitis, 2.917 (2.67, 3.187); and bacterial encephalitis, 4.963 (3.299, 7.466).

Comparative analysis with non-encephalitis acute conditions:

To determine if the observed dementia risk is unique to encephalitis (parenchymal inflammation) or shared with other forms of acute neurological or systemic insults, we compared encephalitis cohort against 3 separate matched cohorts (Sepsis, meningitis and stroke) for dementia, mortality and epilepsy outcomes (Supplementary Table 1). After matching, the encephalitis cohort comprised $n = 4343$ (vs meningitis), $n = 2955$ (vs sepsis) and $n = 3302$ (vs stroke) patients, respectively. The detailed baseline characteristics of each cohort before and after propensity score matching are outlined in Supplementary Table 4.

Encephalitis vs meningitis

Meningitis is an inflammation of meninges, commonly caused by microbial infections and leads to neurological complications [69]. We asked if inflammation in brain parenchymal regions versus meninges influences dementia risk differentially. Interestingly, patients with encephalitis showed a significantly higher risk of developing dementia and related diagnoses (Supplementary Fig. 9) and all individual diagnostic categories compared to patients who had meningitis (Fig. 5A). The RR 95%CI were as follows: Dementia and related diagnoses, 1.804 (1.719, 1.893) (Supplementary Fig. 9); Alzheimer's disease, 1.257 (1.088, 1.453); other degenerative diseases of nervous system, 1.928 (1.805, 2.059); vascular dementia,

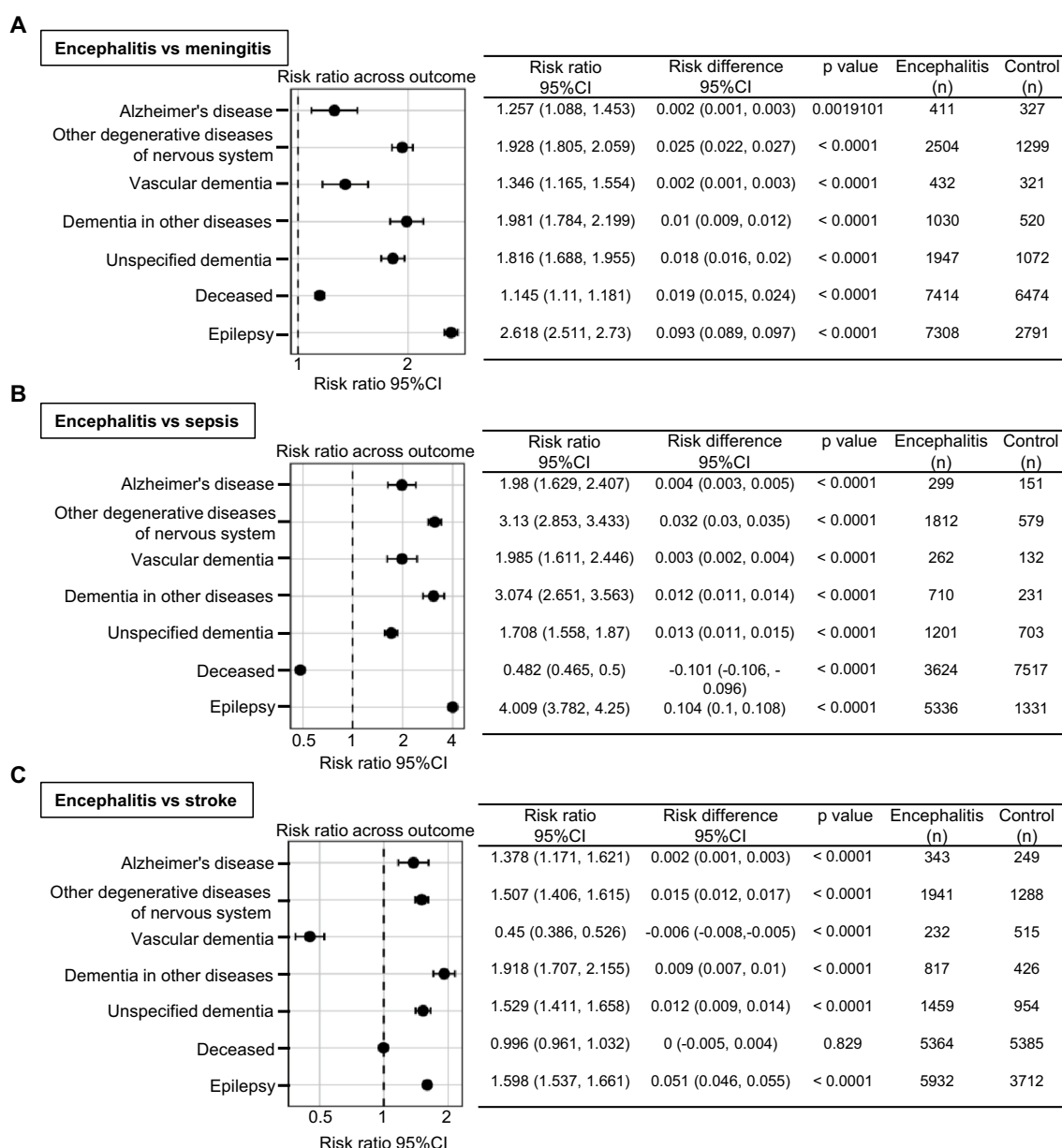


Fig. 5 Risk across individual diagnostic categories of dementia and related diagnoses following encephalitis compared with meningitis, sepsis and stroke. Forest plot illustrating risk of individual diagnostic categories of dementia and related diagnoses following **A** encephalitis vs meningitis **B** encephalitis vs sepsis, and **C** encephalitis vs stroke.

Dashed vertical line represents risk ratio = 1, indicating no significant association for dementia risk among groups. The table lists risk ratios with 95% CI, risk differences with 95% CI, p value and number of dementia patients in each cohort

1.346 (1.165, 1.554); dementia in other diseases, 1.981 (1.784, 2.199); unspecified dementia, 1.816 (1.688, 1.955) (Fig. 5A). The all-cause mortality (RR 95% CI 1.145 (1.11, 1.181)) and epilepsy (RR 95% CI 2.618 (2.511, 2.73)) outcome risks were also higher in the encephalitis cohort (Fig. 5A). This suggests that direct brain parenchymal inflammation is a stronger risk factor for long-term neurodegeneration compared to inflammation confined to meninges.

Encephalitis vs sepsis

Sepsis is an acute condition caused by dysregulated host response to infection. It results in systemic inflammation, organ dysfunction and is often life threatening. Studies suggest that patients with sepsis may suffer from brain dysfunction and are at higher risk of cognitive decline [70–73]. We compared risk of dementia between patients with encephalitis and those with sepsis to assess if direct

brain inflammation confers greater risk than caused by systemic infection. We observed elevated risk of dementia and related diagnoses (Supplementary Fig. 9) and all individual diagnostic categories in encephalitis compared to sepsis cohort (Fig. 5B). The RR 95%CI were as follows: Dementia and related diagnoses, 2.225 (2.089, 2.37) (Supplementary Fig. 9); Alzheimer's disease, 1.98 (1.629, 2.407); other degenerative diseases of nervous system, 3.13 (2.853, 3.433); vascular dementia, 1.985 (1.611, 2.446); dementia in other diseases, 3.074 (2.651, 3.563); and unspecified dementia, 1.708 (1.558, 1.87) (Fig. 5B). This suggests that brain inflammation confers greater risk compared to systemic inflammation in driving cognitive decline. All-cause mortality risk was lower in the encephalitis cohort (RR 95% CI 0.482 (0.465, 0.5)) compared to sepsis. However, epilepsy outcome risk was higher in the encephalitis cohort (RR 95% CI 4.009 (3.782, 4.25)) (Fig. 5B).

Encephalitis vs stroke

Stroke is cerebrovascular event resulting from interruption of blood flow to a region of the brain. Survivors of strokes are at higher risk of developing dementia, particularly vascular dementia [39–42]. Therefore, we compared risk of dementia after encephalitis and vascular/structural brain injury. Encephalitis patients showed higher risk of developing dementia and related diagnoses (Supplementary Fig. 9). Moreover, all individual diagnostic categories were at higher risk in encephalitis cohort, except for vascular dementia (Fig. 5C). The higher incidence of vascular dementia in stroke cohort aligns with the published studies [39–42] and serve as a further internal validity check supporting our cohort construction. The RR 95%CI were as follows: Dementia and related diagnoses, 1.362 (1.294, 1.432) (Supplementary Fig. 9); Alzheimer's disease, 1.378 (1.171, 1.621); other degenerative diseases of nervous system, 1.507 (1.406, 1.615); vascular dementia, 0.45 (0.386, 0.526); dementia in other diseases, 1.918 (1.707, 2.155); and unspecified dementia, 1.529 (1.411, 1.658) (Fig. 5C). Encephalitis did not confer a significant risk for all-cause mortality outcome (RR 95% CI 0.996 (0.961, 1.032) compared to stroke cohort (Fig. 5C). However, as expected, epilepsy (RR 95% CI 1.598 (1.537, 1.661)) outcome risk was higher in the encephalitis cohort (Fig. 5C).

Discussion

In this retrospective cohort study utilizing the TriNetX data, we demonstrate a robust association between encephalitis and subsequent risk for development of dementia. The incidence and prevalence of encephalitis based upon TriNetX database were higher in older adults, reflecting the increased

vulnerability of aging population to inflammatory and infectious neurological conditions. Our findings suggest that encephalitis survivors face an elevated risk of developing dementia, with this risk varies depending on age, encephalitis subtype and individual diagnostic category of dementia and related diagnoses.

Across all age groups, patients with encephalitis had higher incidence of dementia and related diagnoses compared with controls. Subgroup analysis of individual diagnostic categories of dementia and related diagnoses showed varied risks of dementia depending upon the age at which individuals encounter encephalitis. We found that in younger cohorts (< 20 year and 20–40 years), disorders coded under G31 (other degenerative diseases of nervous system) were significantly increased. The lack of association with other dementia subtypes, including Alzheimer's disease, could be due to insufficient follow-up times as these diseases typically develop later in life. Because G31 (other degenerative diseases of nervous system) is a broad category that also includes diagnoses other than dementia, the elevated signal in younger cohorts may reflect long-term post encephalitic neurological sequelae rather than dementia and its long-term clinical significance remains to be determined. In contrast, older adults (40–60 years and > 60 years) showed significant elevated risk across all dementia categories and aligns with the recent studies that age is a strong risk factor for developing dementia. This association also persisted when G31 (other degenerative diseases of nervous system) was excluded in the sensitivity analysis. These data suggest that while encephalitis is associated with dementia and related diagnoses across lifespan, the nature of this association is age-dependent with a robust and specific dementia risk in older adults.

We observed differential risk profile in dementia based upon encephalitis etiology. Both non-infectious/post-infectious inflammatory and viral encephalitis were associated with significantly increased risk of dementia and related diagnoses and all individual diagnostic categories, with non-infectious/post-infectious inflammatory encephalitis exhibiting the highest relative risk. Due to limitations of ICD-10 coding, we could not differentiate the long-term cognitive impacts of specific subtypes such as autoantibody-mediated encephalitis (anti-NMDA receptor encephalitis) and autoimmune variant. Nonetheless, this high risk observed aligns with previous studies showing that autoimmune encephalitis involves dysregulated inflammation targeting neural antigens [59, 61, 62]. This chronic and sustained neuroinflammation may accelerate the development of neurodegenerative sequelae and promote dementia. Viral encephalitis also showed elevated risks supporting prior work that herpes simplex virus and other viruses may contribute to long-term cognitive impairment [26, 46, 47, 65]. Unspecified encephalitis and fungal/parasitic

encephalitis were also associated with increased risk of dementia and related diagnoses and most individual diagnostic categories. Surprisingly, bacterial encephalitis is the only subtype that did not show any significant association with dementia and related diagnoses. While bacterial infections that result in encephalitis are severe, a lower number of encephalitis cases ($n = 1849$) might be contributing to a lack of association. Alternatively, bacterial infections are managed by aggressive antimicrobial and anti-inflammatory therapies that may fail to trigger sustained neuroinflammatory cascade required for progressive neurodegeneration.

The use of comparator cohorts further supports the specificity of our findings. Our data demonstrated that dementia risk following encephalitis is significantly higher in encephalitis than observed in patients with sepsis, meningitis or stroke. As expected, stroke showed the strongest association with vascular dementia providing internal validation. In sepsis, inflammation is systemic, in meningitis it is primarily restricted to meninges while in stroke it results in vascular and structural injury. However, none of them involve the degree of parenchymal injury/inflammation observed in encephalitis, where neuronal integrity is affected within cerebral tissue itself. This suggests that inflammation and direct injury to the brain parenchyma results in a higher risk and is a critical factor driving subsequent long-term cognitive decline. This also aligns with the studies on traumatic brain injury, where a single episode of brain tissue damage can lead to persistent neuroinflammation, white matter degeneration and Alzheimer's Disease like neuropathological changes years after the acute event [43, 74].

Beyond the immediate clinical implications for encephalitis survivors, these findings raise important questions about how acute central nervous system inflammation may influences long-term neurological sequelae. While our study does not offer any mechanistic insights, the observed association suggests that encephalitis could serve as a useful epidemiological framework for studying the long-term cognitive impact of acute neuroinflammatory insults. Chronic neuroinflammation is increasingly recognized as a key driver in Alzheimer disease, vascular dementia and other neurodegenerative disorders [6–8]. Although encephalitis affects a smaller population than chronic neurodegenerative diseases, these cases represent a well-defined inflammatory insult to the central nervous system, allowing assessment of how acute and persistent inflammation impacts neuronal integrity and function. Our findings that non-infectious/post-infectious inflammatory and viral encephalitis conditions confer the highest relative risk of subsequent dementia further supports the hypothesis that inflammatory processes may accelerate cognitive vulnerability. Future studies may focus on follow up of encephalitis survivors to better characterize the timing and trajectory of cognitive decline. Moreover, integration of

genetic and biomarker data may provide unique mechanistic insights into how inflammation promotes dementia.

Limitations

This study has several strengths as we utilized large real world electronic health records enhancing generalizations across diverse clinical settings. We also examined multiple dementia individual diagnostic categories, age groups, encephalitis etiologies and comparator cohorts, providing a comprehensive evaluation. Moreover, we included epilepsy as a positive control outcome, that yielded positive associations across all comparisons, further strengthened our findings [50, 67, 68]. However, there are several limitations of our study. We have used diagnostic ICD-10 codes, which may misclassify encephalitis and dementia subtypes due to bias and inconsistent coding practices. We were unable to incorporate radiological or laboratory data such as magnetic resonance imaging or cerebrospinal fluid (CSF) analyses to independently verify diagnoses or assess disease severity. This limitation is particularly relevant for encephalitis subtypes based on etiology which we could not validate against antibody, CSF or imaging data. Unspecified encephalitis was the largest subtype, suggesting that a substantial proportion of cases could not be etiologically characterized in electronic health records, which may affect subtype-specific risk estimates. Another limitation is the potential for surveillance bias. Patients surviving encephalitis typically undergo more frequent clinical follow-up than matched controls, making subsequent cognitive symptoms more likely to be detected and diagnosed. To mitigate the potential influence of acute surveillance bias, we conducted landmark analyses excluding dementia outcomes diagnosed within 3, 6 and 12 months post-encephalitis. The dementia risk remained significantly elevated in older adults. Furthermore, our comparator cohorts (sepsis, meningitis and stroke), which also receives frequent follow up showed lower dementia risk than the encephalitis cohort, suggesting that the long-term association is not solely driven by early clinical surveillance. Although we have used propensity score matching to reduce confounding variables, other factors such as disease severity, lifestyle, and genetics could not be excluded. Moreover, dementia may be undiagnosed in younger population due to shorter follow-up period. In addition, given the number of age groups, etiological subtypes, individual diagnostic categories and comparator cohorts examined, we could not apply formal corrections for multiple comparisons as the TriNetX analytics does not support automated false-discovery rate correction on propensity-matched outputs. We therefore pre-specified dementia and related diagnoses in the overall cohort as our primary endpoint, while all other analyses

as secondary and exploratory, and report all associations with effect estimates and 95% confidence intervals. Furthermore, high mortality associated with severe infections represents a substantial competing risk for long-term dementia and related outcomes. Although competing risks and composite analyses were performed to address this, survival bias may still influence the risk estimates in our analysis. Finally, since this is a retrospective study, we can only establish an association. Definitive proof will require a combination of prospective and mechanistic studies in the future.

Conclusions

This large-scale cohort study demonstrates that encephalitis is associated with a significantly increased risk of dementia, with variations in risk by age and encephalitis etiology. These findings highlight the importance of long-term cognitive monitoring for encephalitis survivors across all age groups and support the use of encephalitis as a model to study acute brain inflammation driven neurodegeneration. Understanding the mechanisms linking central nervous system inflammation to neuronal injury may inform therapeutic strategies aimed at preventing and slowing neurodegenerative disease progression.

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Author contributions PS, GG and Aditi designed the initial study which was updated and refined with contributions from TB, XF, MM, SS and AL. Aditi wrote the first draft of the manuscript. All authors revised and contributed to writing the manuscript. All authors approve the final version of the manuscript and accept accountabilities for all aspects of the work reported in the manuscript.

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Data availability The data analyzed in this study were obtained from US Collaborative Network within the TriNetX Research Network. The data that support the findings of this study are available from TriNetX but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available.

Data are however, available from the authors upon reasonable request and with permission of the TriNetX Research Network.

Declarations

Conflicts of interest The authors declare no competing interests.

Ethical approval and consent to participate This study was conducted in accordance with the ethical principles of the Declaration of Helsinki. Data for this analysis was from the TriNetX database, which is a HIPAA, GDPR, and LGPD compliant platform of federated, deidentified, and anonymous health records. As all data were fully deidentified and anonymous, this study was determined to be exempt from IRB review and the requirement for informed consent was waived.

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